

Advanced Python Programming(lab practical)

6th Program

1.GUI Based Calculator

Program:

```
from tkinter import *

def calculate(op):

    num1 = float(entry1.get())
    num2 = float(entry2.get())

    if op == "+":
        result = num1 + num2
    elif op == "-":
        result = num1 - num2
    elif op == "*":
        result = num1 * num2
    elif op == "/":
        if num2 != 0:
            result = num1 / num2
        else:
            result = "Cannot divide by zero"

    lbl_result.config(text="Result = " + str(result))

root = Tk()
root.title("Simple Calculator")
```

```
root.geometry("300x250")
```

```
Label(root, text="First Number").pack()
```

```
entry1 = Entry(root)
```

```
entry1.pack()
```

```
Label(root, text="Second Number").pack()
```

```
entry2 = Entry(root)
```

```
entry2.pack()
```

```
Button(root, text="Add", width=10,  
        command=lambda: calculate("+")).pack()
```

```
Button(root, text="Subtract", width=10,  
        command=lambda: calculate("-")).pack()
```

```
Button(root, text="Multiply", width=10,  
        command=lambda: calculate("*")).pack()
```

```
Button(root, text="Divide", width=10,  
        command=lambda: calculate("/")).pack()
```

```
lbl_result = Label(root, text="Result = ")
```

```
lbl_result.pack()
```

```
root.mainloop()
```

Output:

The image shows a web-based calculator interface. It consists of the following elements:

- A label "First Number" above a text input field containing the value "27".
- A label "Second Number" above a text input field containing the value "30".
- A vertical stack of four buttons: "Add", "Subtract", "Multiply", and "Divide".
- A label "Result = 57.0" displayed below the buttons.

2. Student attendance dashboard

Program:

```
import tkinter as tk
from tkinter import messagebox

def calculate():
    try:
        total = int(total_entry.get())
        present = int(present_entry.get())

        if present > total or total <= 0:
            raise ValueError

        percentage = (present / total) * 100
        status = "Eligible" if percentage >= 75 else "Attendance Shortage"

        result.config(
            text=f"Attendance: {percentage:.2f}%\nStatus: {status}"
        )
    except ValueError:
        messagebox.showerror("Error", "Enter valid attendance details")

root = tk.Tk()
root.title("Student Attendance Dashboard")
root.geometry("420x350")
root.resizable(False, False)
```

```
tk.Label(root, text="STUDENT ATTENDANCE DASHBOARD",  
        font=("Arial", 16, "bold")).pack(pady=25)
```

```
frame = tk.Frame(root)  
frame.pack(pady=10)
```

```
tk.Label(frame, text="Student Name").grid(row=0, column=0, padx=10, pady=10)  
name_entry = tk.Entry(frame, width=25)  
name_entry.grid(row=0, column=1)
```

```
tk.Label(frame, text="Total Classes").grid(row=1, column=0, padx=10, pady=10)  
total_entry = tk.Entry(frame, width=25)  
total_entry.grid(row=1, column=1)
```

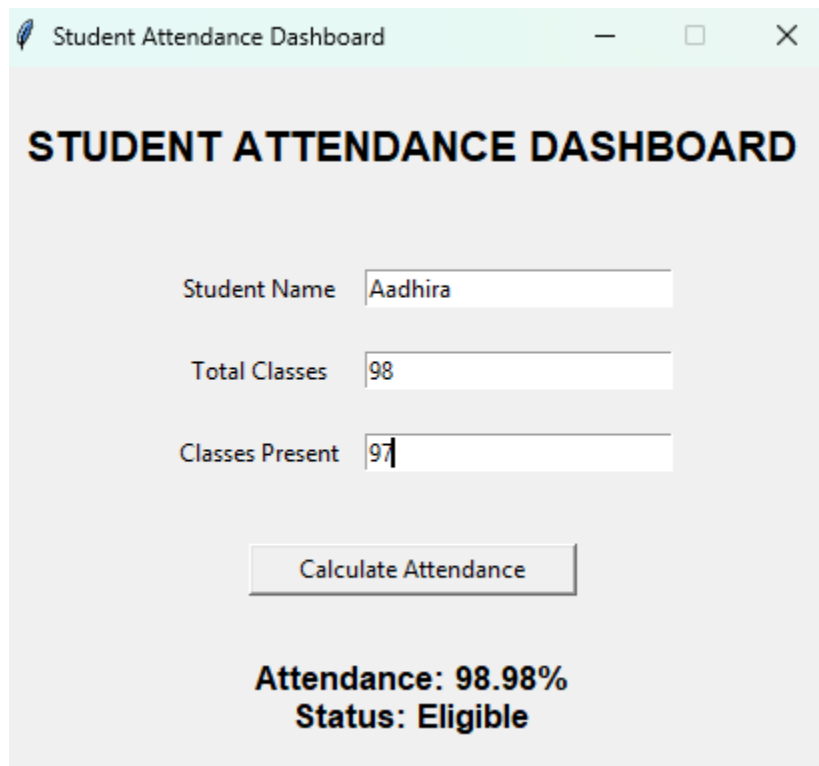
```
tk.Label(frame, text="Classes Present").grid(row=2, column=0, padx=10, pady=10)  
present_entry = tk.Entry(frame, width=25)  
present_entry.grid(row=2, column=1)
```

```
tk.Button(root, text="Calculate Attendance",  
         command=calculate, width=22).pack(pady=15)
```

```
result = tk.Label(root, text="", font=("Arial", 13, "bold"))  
result.pack(pady=15)
```

```
root.mainloop()
```

Output:



The screenshot shows a web browser window titled "Student Attendance Dashboard". The page has a light gray background and a white title bar. The main heading is "STUDENT ATTENDANCE DASHBOARD" in bold black text. Below the heading, there are three input fields with labels: "Student Name" (containing "Aadhira"), "Total Classes" (containing "98"), and "Classes Present" (containing "97"). A "Calculate Attendance" button is positioned below these fields. At the bottom, the calculated attendance is displayed as "Attendance: 98.98%" and the status as "Status: Eligible".

Field	Value
Student Name	Aadhira
Total Classes	98
Classes Present	97

Attendance: 98.98%
Status: Eligible

3. Personal expense tracker

Program:

```
import tkinter as tk

from tkinter import messagebox

total = 0

def add_expense():

    global total

    try:

        amount = float(amount_entry.get())

        if amount <= 0:

            raise ValueError

        category = category_entry.get()

        if not category:

            messagebox.showwarning("Warning", "Enter expense category")

            return

        total += amount

        expense_list.insert(tk.END, f'{category} - ₹{amount:.2f}')

        total_label.config(text=f'Total Expense: ₹ {total:.2f}')

        amount_entry.delete(0, tk.END)

        category_entry.delete(0, tk.END)

    except ValueError:

        messagebox.showerror("Error", "Enter a valid amount")

root = tk.Tk()

root.title("Personal Expense Tracker")

root.geometry("450x450")
```

```
root.resizable(False, False)
```

```
tk.Label(root, text="PERSONAL EXPENSE TRACKER",  
         font=("Arial", 17, "bold")).pack(pady=20)
```

```
tk.Label(root, text="Category").pack()  
category_entry = tk.Entry(root, width=35)  
category_entry.pack(pady=5)
```

```
tk.Label(root, text="Amount").pack()  
amount_entry = tk.Entry(root, width=35)  
amount_entry.pack(pady=5)
```

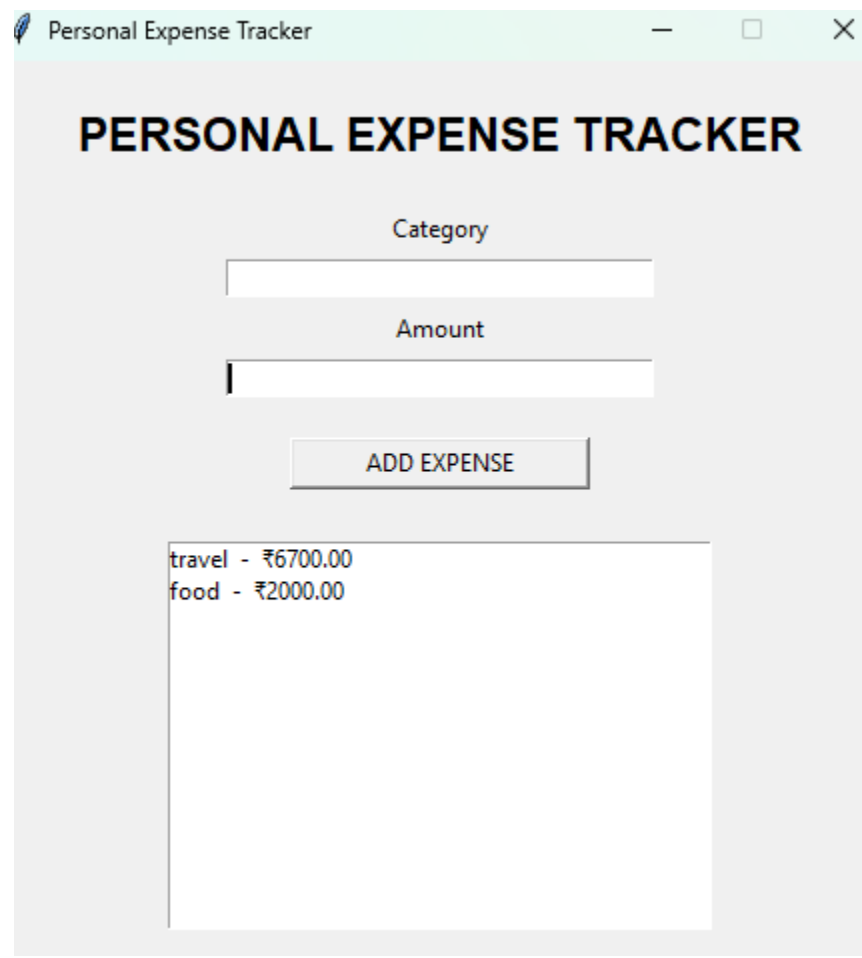
```
tk.Button(root, text="ADD EXPENSE",  
          command=add_expense, width=20).pack(pady=15)
```

```
expense_list = tk.Listbox(root, width=45, height=12)  
expense_list.pack(pady=10)
```

```
total_label = tk.Label(root, text="Total Expense: ₹0.00",  
                        font=("Arial", 14, "bold"))  
total_label.pack(pady=15)
```

```
root.mainloop()
```

Output:



The screenshot shows a web application window titled "Personal Expense Tracker". The main heading is "PERSONAL EXPENSE TRACKER". Below the heading, there are two input fields: "Category" and "Amount". A button labeled "ADD EXPENSE" is positioned below the "Amount" field. At the bottom, a list of expenses is displayed: "travel - ₹6700.00" and "food - ₹2000.00".

Personal Expense Tracker

PERSONAL EXPENSE TRACKER

Category

Amount

ADD EXPENSE

travel - ₹6700.00
food - ₹2000.00

4. Digital library book manager

Program:

```
import tkinter as tk

from tkinter import messagebox

def add_book():

    book_id = id_entry.get()

    title = title_entry.get()

    author = author_entry.get()

    if not book_id or not title or not author:

        messagebox.showwarning("Warning", "Fill all fields")

        return

    books.insert(tk.END, f'{book_id} | {title} | {author} | Available')

    id_entry.delete(0, tk.END)

    title_entry.delete(0, tk.END)

    author_entry.delete(0, tk.END)

def delete_book():

    try:

        books.delete(books.curselection())

    except:

        messagebox.showwarning("Warning", "Select a book to delete")

root = tk.Tk()

root.title("Digital Library Manager")

root.geometry("600x450")

root.resizable(False, False)
```

```
tk.Label(root, text="DIGITAL LIBRARY BOOK MANAGER",  
         font=("Arial", 18, "bold")).pack(pady=20)
```

```
frame = tk.Frame(root)  
frame.pack()
```

```
tk.Label(frame, text="Book ID").grid(row=0, column=0, padx=10, pady=8)  
id_entry = tk.Entry(frame, width=30)  
id_entry.grid(row=0, column=1)
```

```
tk.Label(frame, text="Book Title").grid(row=1, column=0, padx=10, pady=8)  
title_entry = tk.Entry(frame, width=30)  
title_entry.grid(row=1, column=1)
```

```
tk.Label(frame, text="Author").grid(row=2, column=0, padx=10, pady=8)  
author_entry = tk.Entry(frame, width=30)  
author_entry.grid(row=2, column=1)
```

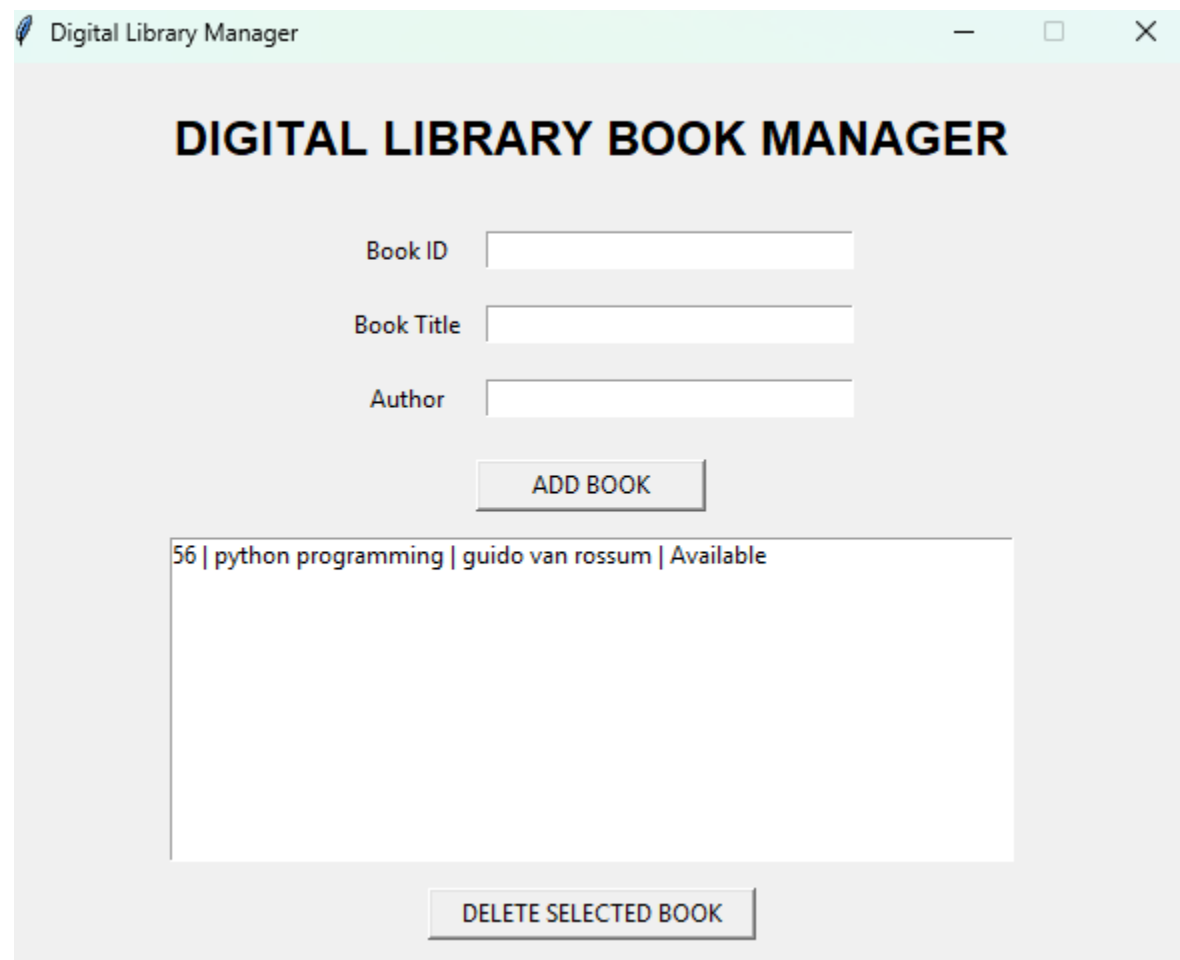
```
tk.Button(root, text="ADD BOOK",  
          command=add_book, width=15).pack(pady=12)
```

```
books = tk.Listbox(root, width=70, height=10)  
books.pack()
```

```
tk.Button(root, text="DELETE SELECTED BOOK",  
          command=delete_book, width=22).pack(pady=12)
```

```
root.mainloop()
```

Output:



The image shows a screenshot of a web application titled "Digital Library Manager". The application has a light green header bar with the title and standard window controls (minimize, maximize, close). Below the header, the main content area has a light gray background. At the top of this area is the title "DIGITAL LIBRARY BOOK MANAGER" in bold black text. Below the title are three input fields labeled "Book ID", "Book Title", and "Author". Below these fields is a button labeled "ADD BOOK". Below the button is a large white rectangular area containing a single line of text: "56 | python programming | guido van rossum | Available". At the bottom of the main content area is a button labeled "DELETE SELECTED BOOK".

Digital Library Manager

DIGITAL LIBRARY BOOK MANAGER

Book ID

Book Title

Author

ADD BOOK

56 | python programming | guido van rossum | Available

DELETE SELECTED BOOK

5. Study planner and Task manager

Program:

```
import tkinter as tk
```

```
from tkinter import messagebox
```

```
def add_task():
```

```
    task = task_entry.get()
```

```
    priority = priority_var.get()
```

```
    if not task:
```

```
        messagebox.showwarning("Warning", "Enter a task")
```

```
    return
```

```
    tasks.insert(tk.END, f"[{priority}] {task}")
```

```
    task_entry.delete(0, tk.END)
```

```
def complete_task():
```

```
    try:
```

```
        index = tasks.curselection()[0]
```

```
        text = tasks.get(index)
```

```
        if "✓" not in text:
```

```
            tasks.delete(index)
```

```
            tasks.insert(index, "✓ " + text)
```

```
    except:
```

```
        messagebox.showwarning("Warning", "Select a task")
```

```
def delete_task():  
    try:  
        tasks.delete(tasks.curselection())  
    except:  
        messagebox.showwarning("Warning", "Select a task")
```

```
root = tk.Tk()  
root.title("Study Planner & Task Manager")  
root.geometry("520x500")  
root.resizable(False, False)
```

```
tk.Label(root, text="STUDY PLANNER",  
         font=("Arial", 20, "bold")).pack(pady=20)
```

```
tk.Label(root, text="Enter Study Task").pack()
```

```
task_entry = tk.Entry(root, width=42)  
task_entry.pack(pady=8)
```

```
tk.Label(root, text="Priority").pack()
```

```
priority_var = tk.StringVar(value="Medium")
```

```
priority = tk.OptionMenu(  
    root, priority_var, "High", "Medium", "Low"  
)  
priority.pack(pady=8)
```

```
tk.Button(root, text="ADD TASK",
          command=add_task, width=18).pack(pady=10)

tasks = tk.Listbox(root, width=55, height=13)
tasks.pack(pady=10)

button_frame = tk.Frame(root)
button_frame.pack()

tk.Button(button_frame, text="✓ Complete",
          command=complete_task, width=14).grid(row=0, column=0, padx=5)

tk.Button(button_frame, text="Delete",
          command=delete_task, width=14).grid(row=0, column=1, padx=5)

root.mainloop()
```

Output:

 Study Planner & Task Manager—□×

STUDY PLANNER

Enter Study Task

Priority

Low 

ADD TASK

[Medium] problem solving

[High] python

[Low] value education

✓ Complete

Delete